

ITM SLS Baroda University
SoCSET
BTech Semester III (Batch 2025-29) CET II Syllabus
Data Structure and Algorithms

1. Queue Data Structure

- a. What is Queue?
- b. Importance of Queue
- c. Types of Queue
 - i. Simple/Linear Queue
 - ii. Circular Queue
 - iii. Double Ended Queue
 - iv. Priority Queue
- d. Operations on all Types of Queues
 - i. Insertion into Queue
 - ii. Deletion from Queue
 - iii. Traversing in a Queue
 - iv. Peeking into a Queue
- e. Computer Applications of all Types of Queues

2. Linked List Data Structure

- a. What is Linked List?
- b. Difference between Arrays and Linked List
- c. Importance of Linked Lists
- d. Types of Linked Lists
 - i. Singly Linked List
 - ii. Doubly Linked List
 - iii. Circular Linked List
- e. Operations on all Types of Linked List
 - i. Insertion into Singly Linked List, Doubly Linked List and Circular Linked List
 - ii. Deletion from Singly Linked List, Doubly Linked List and Circular Linked List
 - iii. Traversing in Singly Linked List, Doubly Linked List and Circular Linked List
- f. Computer Applications of all Types of Linked List
